



Lecture Notes

Institute Name:	Silver Oak College of Computer Application		
Programme Name:	Master of Science Data Science	Semester:	1
Course Code:	DSM5061C	Course Name:	Information and Data Security
Unit No.:	3	Unit Name:	Security Technologies for Data-Driven Systems

1. Introduction to Security Technologies for Data-Driven Systems

1.1 Meaning of a Data-Driven System

A **data-driven system** is a system in which data is collected, stored, processed, analyzed, and used to support operational or business decisions.

Examples:

Banking systems

E-commerce platforms

Healthcare information systems

Social media platforms

Cloud-based applications

Educational management systems

IoT platforms

Artificial Intelligence and Machine Learning systems

Big Data analytics platforms

Basic Data-Driven System

Data Sources



Data Collection





Data Storage



Data Processing



Data Analysis



Decision / Output

Security must be maintained throughout this entire lifecycle.

1.2 Why Security is Required

A data-driven system may contain:

Personal information

Financial information

Authentication credentials

Business information

Intellectual property

Customer records

Medical information

Machine learning datasets

AI models

If this information is compromised, consequences can include:

Data leakage

Financial loss

Identity theft

Privacy violations

Service disruption

Reputation damage

Legal and regulatory consequences

Incorrect business decisions

2. Security Objectives

A fundamental security model is the **CIA Triad**.

Security Objective	Meaning	Example
Confidentiality	Prevent unauthorized access	Only authorized employees access customer data
Integrity	Prevent unauthorized modification	Transaction amount cannot be changed illegally
Availability	Ensure authorized users can access services	Banking service remains available

2.1 Confidentiality

Confidentiality ensures that information is accessible only to authorized entities.

Example:

A university stores student examination results. Only authorized faculty members should access them.

Common mechanisms:

Encryption

Authentication

Access control

VPN

Password protection

2.2 Integrity

Integrity ensures that data remains accurate and has not been improperly modified.

Example:

Suppose a student's original marks are:



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Database:

Student = Rahul

Marks = 78

An attacker changes them to:

Marks = 98

The confidentiality of the record may remain intact, but its **integrity has been compromised**.

Methods for protecting integrity:

Hashing

Digital signatures

Access controls

Audit logs

Checksums

2.3 Availability

Availability ensures that legitimate users can access systems and information when required.

Example:

A university examination portal must remain available during online examinations.

Threats to availability include:

Denial-of-Service attacks

Distributed Denial-of-Service attacks

Hardware failures

Network failures

Ransomware

Cloud service failures



PART A — NETWORK AND COMMUNICATION SECURITY

3. Network Security Fundamentals

3.1 Definition

Network security refers to the policies, technologies, controls, and practices used to protect computer networks, devices, communication channels, and transmitted data from unauthorized access, misuse, modification, disruption, or destruction.

A network may include:

Users



Computers



Switches



Routers



Internet



Cloud / Server

Every communication point creates a potential security risk.

3.2 Goals of Network Security

Network security attempts to provide:

Confidentiality

Integrity

Availability

Authentication

Authorization

Accountability

Authentication



Authentication answers:

"Who are you?"

Examples:

Username and password

OTP

Fingerprint

Face recognition

Security token

Authorization

Authorization answers:

"What are you allowed to access?"

For example:

User → Login



Authentication



Authorization



Access permitted resources

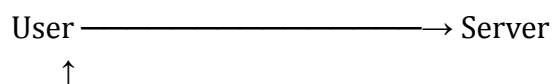
A user may successfully authenticate but still not have permission to access a particular database.

4. Common Network Security Threats

4.1 Eavesdropping

An attacker secretly observes communication.

Example:





Attacker

If communication is not encrypted, sensitive information may be exposed.

4.2 Man-in-the-Middle Attack

In a conceptual MITM attack, an attacker positions themselves between two communicating parties.

User ←————→ Attacker ←————→ Server

The attacker may attempt to intercept or manipulate communication.

Protection mechanisms include:

TLS

HTTPS

VPN

Certificate validation

4.3 Denial-of-Service

A DoS attack attempts to make a service unavailable to legitimate users.

Large amount of malicious traffic



Web Server



Resources exhausted



Legitimate users
cannot access

4.4 Spoofing

Spoofing involves pretending to be another device, user, service, or identity.

Examples:

IP spoofing

Email spoofing



DNS spoofing

Website spoofing

4.5 Malware

Malicious software designed to compromise systems.

Examples:

Virus

Worm

Trojan

Ransomware

Spyware

5. Network Security Controls

Security controls can be divided into several categories.

Preventive Controls

Designed to prevent attacks.

Examples:

Firewall

Access control

Encryption

Strong authentication

Detective Controls

Designed to detect suspicious activity.

Examples:

Intrusion Detection System

Log monitoring

Security Information and Event Management

Corrective Controls

Designed to restore systems after an incident.

Examples:

Backup restoration

Incident response

Disaster recovery

6. Firewalls

6.1 Definition

A **firewall** is a security mechanism that monitors and controls network traffic according to predefined security rules.

It can be implemented as:

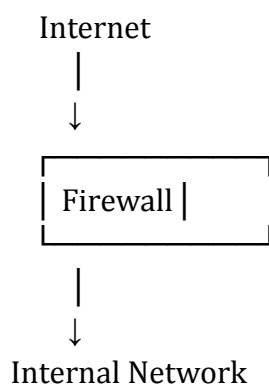
Hardware

Software

Cloud service

Virtual network security component

6.2 Basic Firewall Operation



The firewall examines network traffic and decides whether to:

ALLOW

or

BLOCK

6.3 Firewall Rules

A firewall rule may consider:

Source IP

Destination IP

Port

Protocol

Direction

Application

User identity

Example:

Source	Destination	Port	Protocol	Action
Internal Network	Web Server	443	TCP	Allow
External Network	Database	3306	TCP	Block
Internal Network	DNS Server	53	UDP	Allow

6.4 Types of Firewalls

1. Packet-Filtering Firewall

Examines individual packets.

Parameters may include:

Source IP

Destination IP



Port

Protocol

Advantages:

Simple

Fast

Limitation:

Limited understanding of application-level activity

2. Stateful Firewall

Maintains information about active network connections.

For example:

Client → Server

Connection established



Firewall tracks session



Return traffic recognized

It can make decisions based on the state of a connection.

3. Proxy Firewall

Acts as an intermediary between client and server.

Client



Proxy Firewall



Internet Server

The client may not communicate directly with the destination server.

4. Next-Generation Firewall

NGFWs can combine traditional firewall functionality with capabilities such as:

Application awareness



Intrusion prevention

Advanced traffic inspection

User-based policies

Threat intelligence integration

6.5 Firewall Limitations

A firewall is not a complete security solution.

It may not protect against:

Insider threats

Compromised credentials

Social engineering

Malicious activity inside an allowed connection

Vulnerabilities in legitimate applications

Poorly configured access controls

Important Classroom Point

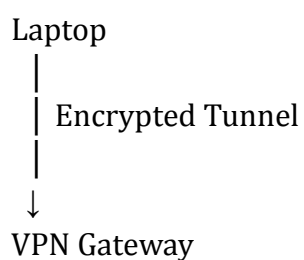
Firewall = one layer of security, not complete security.

7. VPN Concepts

7.1 Definition

A **Virtual Private Network (VPN)** creates a protected communication tunnel over an existing network, commonly the Internet.

Conceptually:





Private Network

7.2 Why VPN is Used

VPNs can provide:

Protected communication

Secure remote access

Privacy of traffic from certain network observers

Secure connectivity between locations

7.3 Remote-Access VPN

Used when an employee wants to securely connect to an organization's network from outside.

Employee



Internet



VPN Gateway



Organization Network

7.4 Site-to-Site VPN

Connects two networks.

Office A



VPN Tunnel



Internet



VPN Tunnel



Office B

This is useful for connecting:



Branch offices

Data centers

Corporate networks

7.5 VPN Security Concepts

Important concepts include:

Encryption

Protects data from unauthorized observation.

Authentication

Verifies the identity of communicating parties.

Tunneling

Encapsulates network traffic for transmission through another network.

Integrity

Helps detect unauthorized modification.

8. Secure Communication

8.1 Meaning

Secure communication means protecting information while it is being transmitted between systems.

Example:

Client —————→ Server
Protected
Communication

8.2 Encryption

Encryption transforms readable data (**plaintext**) into an unreadable form (**ciphertext**).

Plaintext



Encryption



Ciphertext



Decryption



Plaintext

8.3 Symmetric Encryption

The same secret key is used for encryption and decryption.

Same Key



Plaintext → Encryption → Ciphertext



Decryption



Plaintext

Advantages:

Fast

Suitable for large amounts of data

Challenge:

Secure key distribution

8.4 Asymmetric Encryption

Uses a pair of keys:

Public key

Private key

Conceptually:

Public Key → Encryption

Private Key → Decryption

Common uses include:



Secure key exchange

Digital signatures

Authentication

9. TLS and HTTPS

HTTPS is HTTP protected using TLS.

Basic flow:

Browser



HTTPS/TLS



Web Server

TLS provides security properties including:

Encryption

Integrity protection

Server authentication through certificates

Example:

http://example.com

versus

https://example.com

HTTPS is important for:

Online banking

E-commerce

Login systems

University portals

Cloud applications



PART B — CLOUD SECURITY

10. Cloud Computing Fundamentals

10.1 Definition

Cloud computing is a model for providing computing resources such as:

Servers

Storage

Databases

Networking

Applications

Processing power

over a network, typically on demand.

10.2 Cloud Service Models

IaaS — Infrastructure as a Service

Provides infrastructure resources.

Examples:

Virtual machines

Storage

Networking

Customer generally manages:

Operating system

Applications

Data

PaaS — Platform as a Service



Provides a platform for application development and deployment.

The provider manages more of the underlying infrastructure.

SaaS — Software as a Service

Provides complete applications through the cloud.

Examples:

Online office applications

Cloud-based email

CRM systems

10.3 Cloud Deployment Models

Public Cloud

Resources are provided by a cloud provider to multiple customers.

Private Cloud

Cloud infrastructure is dedicated to one organization.

Hybrid Cloud

Combines private and public cloud environments.

Multi-Cloud

An organization uses services from multiple cloud providers.

11. Cloud Security

Cloud security refers to the policies, technologies, processes, and controls used to protect:

Cloud infrastructure

Applications

Data

Users

Virtual machines

APIs

Network resources

12. Shared Responsibility Model

This is one of the **most important concepts in cloud security**.

Cloud security responsibility is divided between:

Cloud Provider

+

Cloud Customer

The exact division depends on the service model and provider.

Conceptual Example

Responsibility Area	Cloud Provider	Customer
Physical data center	✓	
Physical hardware	✓	
Cloud infrastructure	✓	
Operating system	Depends on service	Depends on service
Applications		✓
User accounts		✓
Access permissions		✓
Customer data		✓
Data classification		✓
Secure configuration	Shared / varies	✓

Important Point

Moving data to the cloud does not mean that the organization transfers all security responsibility to the cloud provider.

13. Cloud Security Challenges

13.1 Misconfiguration

Incorrect configuration can expose resources.

Examples:

Publicly accessible storage

Excessive permissions

Open network ports

13.2 Identity and Access Management

Weak credentials or excessive permissions can result in unauthorized access.

Security mechanisms:

Multi-factor authentication

Role-based access control

Least privilege

Strong password policies

13.3 Data Protection

Sensitive cloud data should be protected:

At rest

In transit

During processing where appropriate

13.4 Insecure APIs

Cloud services commonly depend on APIs.

An insecure API may expose:

Data



Authentication functionality

Administrative operations

13.5 Data Loss

Potential causes:

Accidental deletion

Ransomware

Hardware/software failure

Malicious activity

Controls:

Backup

Replication

Recovery procedures

Versioning

PART C — BIG DATA SECURITY

14. Big Data Fundamentals

Big Data generally refers to datasets characterized by large-scale and complex data processing requirements.

Common characteristics are described using the **5 Vs**:

V	Meaning
Volume	Large quantity of data
Velocity	High speed of data generation
Variety	Multiple data formats
Veracity	Data reliability/quality

Value	Useful information obtained from data
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Examples:

Social media data

IoT data

Transaction data

Sensor data

Web logs

15. Security Challenges in Big Data Environments

Big Data environments create security challenges because data may be:

Very large

Distributed

Generated continuously

Stored across multiple systems

Processed by many users

Stored in different formats

15.1 Data Volume

Large volumes make security monitoring more difficult.

Example:

A platform generates millions of log records every day.

The security system must identify meaningful events among huge quantities of data.

15.2 Data Variety

Data may include:

Text

Images

Videos

JSON

Logs

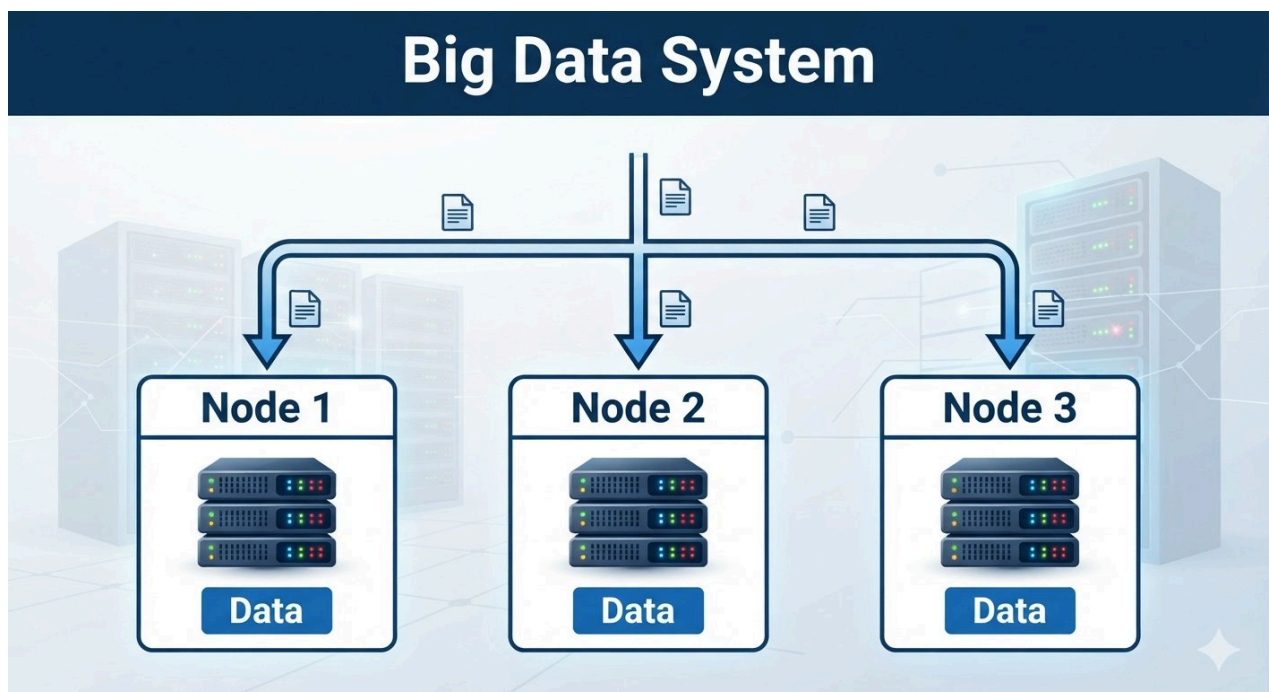
Sensor data

Databases

Each data type may require different security controls.

15.3 Distributed Storage

Big Data systems frequently distribute data across multiple machines.



A security weakness in one component can affect the broader environment.

16. Big Data Security Threats

Common threats include:

Unauthorized Access

Unauthorized users access sensitive datasets.

Data Leakage

Sensitive information is exposed.

Insider Threats

Authorized users misuse their permissions.

Insecure Interfaces

Weak APIs or interfaces expose information.

Data Integrity Attacks

Attackers modify datasets.

Privacy Risks

Combining multiple datasets may reveal sensitive information.

17. Data Access Control

Data access control determines:

Who can access which data, under what conditions, and what operations they can perform.

17.1 Authentication vs Authorization

Authentication	Authorization
Identifies user	Determines permissions
"Who are you?"	"What can you access?"
Login	Access rights

18. Role-Based Access Control

RBAC assigns permissions to roles.

Example:

User





Role



Permissions



Resources

Example:

Role	Student Data	Examination Data	System Configuration
Student	Own records	Own results	No
Faculty	Assigned students	Relevant results	Limited
Administrator	Authorized records	Authorized records	Yes

19. Principle of Least Privilege

A user should receive only the permissions necessary for their legitimate work.

Example:

A data analyst who only needs aggregated sales data does not necessarily need access to customer passwords.

Formula

Required Access



Minimum Necessary Permission



Reduced Attack Surface

20. Data Security in Distributed Systems

Distributed systems store and process data across multiple machines.

Security must therefore protect:

Nodes

Communication

Data

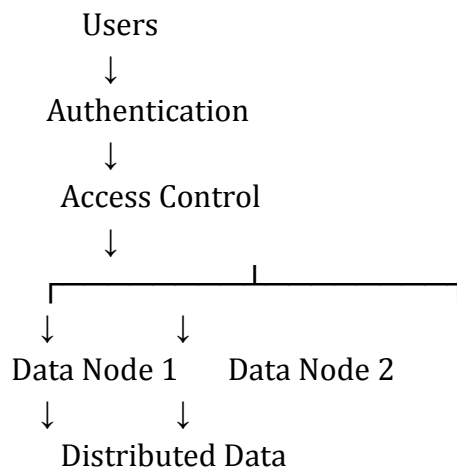
Users

APIs

Credentials

Administrative interfaces

Security Architecture



Important mechanisms:

Encryption

Authentication

Authorization

Secure communication

Audit logging

Backup

Monitoring

PART D — AI AND SECURITY

21. Artificial Intelligence in Cybersecurity

Artificial Intelligence can be used to support cybersecurity operations by processing large volumes of security data.

Traditional security systems often depend heavily on predefined rules.

AI-based approaches can identify patterns from data.

21.1 Traditional Security

Known Rule



Incoming Event



Match Rule?

/ \

Yes No



Alert No Alert

21.2 AI-Assisted Security

Security Data



Feature Extraction



ML / AI Model



Pattern Detection



Risk / Alert

22. Applications of AI in Cybersecurity

AI can support:

Intrusion detection

Malware detection

Spam detection

Phishing detection

Anomaly detection

Fraud detection

User behavior analysis

Security log analysis

Threat intelligence

Automated response

23. AI-Assisted Threat Detection

23.1 Anomaly Detection

Anomaly detection identifies behavior that differs significantly from an established pattern.

Example:

Normal behavior:

Employee:

09:00 → Login

09:15 → Email

10:00 → Database

17:30 → Logout

Sudden behavior:

02:45 → Login

02:50 → Large database download

The system can flag this activity for investigation.

Important Point

An anomaly is **not automatically an attack**.

It is an event that differs from expected behavior.

Human or automated investigation may be required.

24. Machine Learning for Threat Detection

A simplified ML security pipeline:

Security Logs



Data Cleaning



Feature Extraction



Training Dataset



ML Model



Prediction



Normal / Suspicious

Possible algorithms include:

Decision Trees

Random Forest

Support Vector Machines

Neural Networks

Autoencoders

Clustering methods

25. AI-Assisted Malware Detection

AI can learn patterns associated with malicious files or behavior.

Possible input features:

File characteristics

API calls

Network behavior

Process behavior

System activity

Output:

Benign

OR

Suspicious / Malicious

26. Security Automation

Security automation means using software to automatically perform selected security operations.

Example:

Suspicious Event



Detection System



Risk Evaluation



Automated Response



Alert / Block / Isolate

Possible automated actions:

Generate alert

Disable a session

Isolate an endpoint

Block suspicious traffic

Create incident ticket

Collect additional logs

26.1 Advantages

Faster response

Handles large volumes

Reduces repetitive manual tasks

Supports continuous monitoring

26.2 Limitations

False positives

False negatives



Poor-quality training data

Model errors

Automation errors

Attackers adapting to detection systems

Classroom Discussion

Should every AI-generated security alert automatically result in blocking a user?

Students should consider:

False positives

Business impact

Critical systems

Human review

Risk level

PART E — SECURITY OF AI SYSTEMS

This section is extremely important because AI can be both:

A Security Tool

AND

A Security Target

27. Security of AI Systems

AI systems consist of multiple components:

Data



Preprocessing



Training



Model



Deployment

↓
Input

↓
Prediction

Each component can introduce security risks.

28. AI Security Threat Surface

Component	Potential Risk
Training data	Data poisoning
Data pipeline	Unauthorized modification
Model	Model theft
Input	Adversarial manipulation
API	Unauthorized access
Infrastructure	System compromise
Output	Sensitive information leakage

29. Data Poisoning

29.1 Definition

Data poisoning is an attack in which an attacker intentionally manipulates training data so that the resulting AI model learns undesirable or incorrect patterns.

Conceptually:

Original Training Data

+

Malicious Data

↓

Training Process

↓

Compromised Model

29.2 Example



Suppose an image classification model is trained to distinguish:

Cat

Dog

If malicious training examples are deliberately introduced with incorrect labels, the model may learn incorrect relationships.

29.3 Consequences

Data poisoning can cause:

Reduced model accuracy

Incorrect predictions

Specific targeted misclassifications

Loss of trust in the AI system

30. Model Security Risks

An AI model itself can become a security target.

Potential risks include:

Model Theft

An attacker attempts to obtain or reproduce a proprietary model.

Model Inference Risks

An attacker may attempt to learn sensitive information about training data from model behavior.

Unauthorized Model Modification

An attacker changes the model or its deployment environment.

Insecure Model Deployment

A model API may expose sensitive functionality without appropriate authentication or authorization.

31. Adversarial Attacks

31.1 Definition

An adversarial attack attempts to manipulate AI model behavior by providing specially designed inputs.

Conceptually:

Normal Input



AI Model



Correct Prediction

versus

Modified / Adversarial Input



AI Model



Incorrect Prediction

31.2 Conceptual Example

An image classification system identifies:

Image → Cat

A carefully modified input may cause the model to produce:

Image → Incorrect Class

The modification may be difficult for a human observer to notice but significant to the model.

32. Types of Adversarial Attacks — Conceptual

Evasion Attack

Attempts to fool a model during inference.

Training → Normal

Inference → Manipulated Input

Poisoning Attack

Targets training data.

Training Data → Manipulated



Compromised Learning

Model Extraction

Attempts to reproduce model behavior by querying a model.

Membership Inference

Attempts to determine whether a particular record was included in model training.

33. Security Measures for AI Systems

Possible measures include:

Data Validation

Check training data for:

Incorrect labels

Duplicate records

Unexpected patterns

Suspicious sources

Access Control

Restrict:

Training datasets

Model files

Model APIs

Deployment environments

Monitoring

Monitor:



Model performance

Input distributions

Prediction patterns

Unusual API activity

Model Testing

Test models against:

Unexpected inputs

Security scenarios

Robustness conditions

PART F — EMERGING SECURITY CHALLENGES

34. Deepfakes

34.1 Definition

A **deepfake** is synthetic or manipulated media generated or modified using AI techniques to imitate a person's appearance, voice, or behavior.

Types include:

Face manipulation

Face swapping

Synthetic video

Voice cloning

Synthetic images

34.2 Deepfake Security Risks

Deepfakes can contribute to:

Identity impersonation

Fraud

Social engineering

Misinformation

Reputation attacks

Authentication challenges

34.3 Example

Conceptual scenario:

Real Person



AI-based Media Generation



Synthetic Video / Voice



Fake Communication



Victim

The victim may incorrectly believe the communication is genuine.

35. Generative AI

35.1 Definition

Generative AI refers to AI systems capable of generating new content such as:

Text

Images

Audio

Video

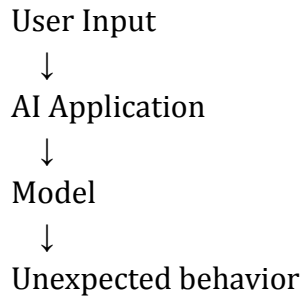
Code

36. Generative AI Security Risks

36.1 Prompt Injection

An attacker may attempt to manipulate an AI system through specially crafted inputs.

Conceptually:



36.2 Sensitive Information Exposure

If an AI application is improperly designed, sensitive information may be exposed through:

Inadequate access controls

Poor data handling

Improper logging

Insecure integrations

36.3 Hallucinated Information

Generative AI can produce plausible-looking but incorrect information.

This becomes a security issue when generated information is used without verification in sensitive environments.

36.4 Malicious Use

Generative AI can potentially lower the effort required to create:

Phishing content

Fraudulent messages

Fake media

Malicious social-engineering content

The security concern is not simply the existence of generative AI, but how it is developed, deployed, accessed, and used.

37. Future Trends in Information Security

The security environment is continuously changing.

Important emerging areas include:

1. Zero Trust Security

Basic principle:

Never automatically trust a user or device merely because it is inside a network.

Conceptual model:

User
↓
Verify Identity
↓
Check Device
↓
Check Context
↓
Authorize Access
↓
Monitor Activity

2. AI-Driven Security Operations

AI can assist security teams in:

Threat detection

Alert prioritization

Log analysis

Incident investigation

3. AI Security

As AI adoption increases, organizations need dedicated controls for:



AI data

AI models

AI APIs

Model access

Training pipelines

4. Cloud-Native Security

As applications increasingly use:

Containers

Microservices

Serverless computing

Cloud APIs

security must operate across dynamic infrastructure.

5. Privacy-Preserving Technologies

Organizations increasingly need methods that allow useful data processing while reducing unnecessary exposure of personal information.

Relevant concepts include:

Data minimization

Encryption

Anonymization

Pseudonymization

Federated learning

Privacy-enhancing technologies

6. Post-Quantum Cryptography

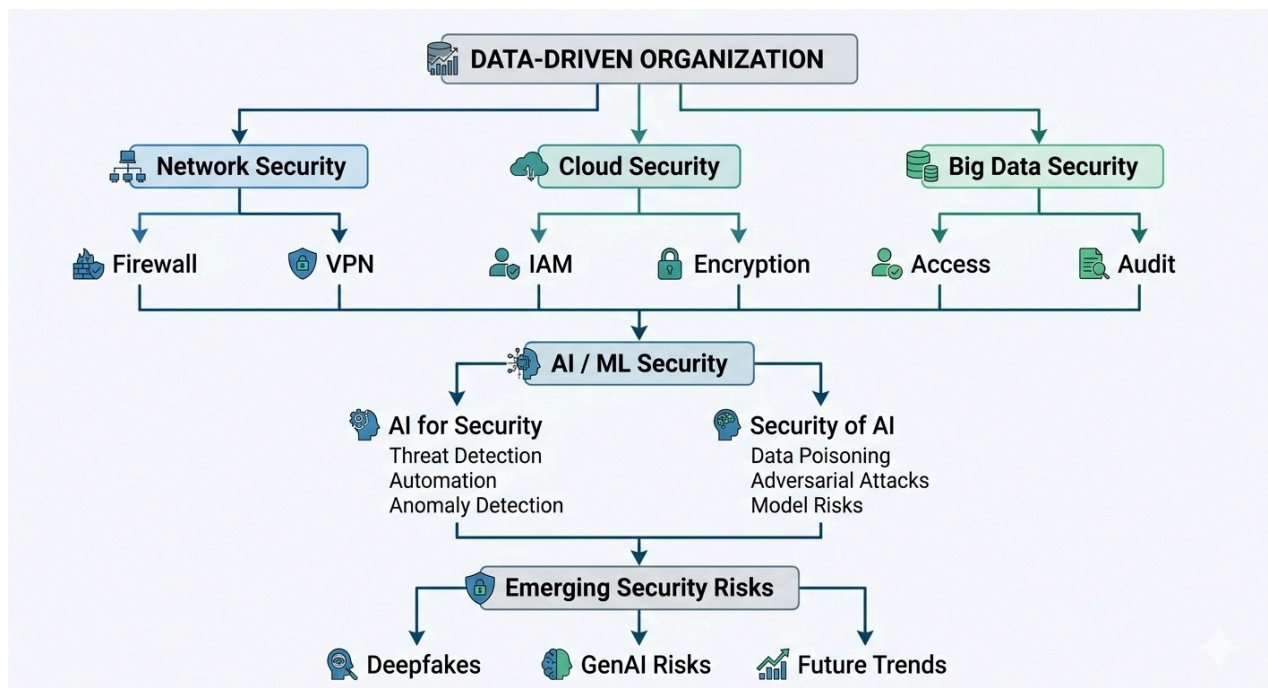
Future cryptographic systems need to consider the possibility of sufficiently capable quantum computers threatening some currently used public-key cryptographic approaches.

Therefore, organizations are researching and adopting **post-quantum cryptographic algorithms**

38. Integrated View of Security Technologies

Students should understand that these technologies are not independent.

A modern data-driven organization may use:



39. Security Across the Data Lifecycle

One of the best ways to explain this unit in class is through the **data lifecycle**.

1. Data Generation
- ↓
2. Data Collection
- ↓
3. Data Transmission
- ↓
4. Data Storage
- ↓
5. Data Processing
- ↓
6. Data Analysis

↓
7. Data Sharing

↓
8. Data Archiving / Deletion

Security should be applied at every stage.

Data Stage	Security Technologies
Generation	Device security, authentication
Collection	Access control, validation
Transmission	TLS, VPN, encryption
Storage	Encryption, IAM
Processing	Access control, secure infrastructure
Analysis	Data governance, privacy controls
Sharing	Authorization, encryption
Archiving	Access control, retention policies
Deletion	Secure deletion

40. End-to-End Example: E-Commerce Organization

Consider an online shopping platform.

Step 1: Customer Login

Security:

Authentication

MFA

HTTPS

Step 2: Network Communication

Security:

Firewall



TLS

Network monitoring

Step 3: Cloud Infrastructure

Security:

IAM

Cloud configuration

Encryption

Shared responsibility controls

Step 4: Customer Database

Security:

Access control

Encryption

Audit logs

Backup

Step 5: Big Data Analytics

Security:

Role-based access

Data classification

Distributed-system security

Step 6: AI Recommendation System

Security:

Training-data protection

Model access control

Model monitoring



Step 7: AI Threat Detection

Security:

Anomaly detection

Automated alerts

Security analytics

Step 8: Emerging Threats

Organization must consider:

Deepfake fraud

Generative AI-based phishing

AI attacks

Cloud threats

Data privacy

41. Important Comparison Tables for Lecture

Firewall vs VPN

Firewall	VPN
Controls network traffic	Creates protected communication tunnel
Mainly traffic filtering/control	Mainly secure connectivity
Can block unauthorized traffic	Protects communication over untrusted networks
Works using security rules	Uses tunneling and cryptographic protection

Authentication vs Authorization

Authentication	Authorization
Verifies identity	Determines permissions
"Who are you?"	"What can you access?"

Username/password	Roles/permissions
Happens before authorization	Follows successful authentication

Traditional Security vs AI-Assisted Security

Traditional Approach	AI-Assisted Approach
Rule-driven	Pattern/data-driven
Known threats	Can identify previously unseen patterns
Manual analysis can be significant	Automation can assist analysis
Fixed rules	Models may adapt based on data
Can be easier to interpret	Some models may be difficult to interpret

Data Poisoning vs Adversarial Attack

Data Poisoning	Adversarial Attack
Primarily targets training process/data	Often targets model inference
Manipulates training data	Manipulates input
Causes model to learn undesirable patterns	Attempts to cause incorrect prediction
Training-stage concern	Inference-stage concern

42. Important Terminology

Students should know these terms before completing the unit:

Term	Meaning
Firewall	Controls network traffic
VPN	Protected network connection/tunnel
Encryption	Converts readable information into protected ciphertext
TLS	Security protocol for protected communication
Cloud Security	Protection of cloud resources and data

IAM	Identity and Access Management
Shared Responsibility	Security responsibilities divided between provider and customer
Big Data	Large and complex datasets
RBAC	Role-Based Access Control
Least Privilege	Minimum required permissions
AI Security	Protection of AI systems
Data Poisoning	Manipulation of training data
Adversarial Attack	Manipulation intended to influence model behavior
Deepfake	AI-generated/manipulated media
Generative AI	AI capable of generating new content
Zero Trust	Continuous verification rather than implicit trust

43. Classroom Case Study

Case: University Data-Driven Learning Platform

A university operates a cloud-based learning platform containing:

Student profiles

Attendance

Examination records

Assignment submissions

Faculty information

Learning analytics

The university also uses AI to identify students who may require academic support.

Security Questions

Which data requires confidentiality?



How can integrity of examination records be maintained?

How can availability of the learning platform be protected?

Where should a firewall be placed?

Where can VPN be used?

Which users should have access to examination records?

How can RBAC be applied?

What is the cloud provider responsible for?

What is the university responsible for?

What could happen if training data for the AI system is poisoned?

How could an adversarial attack affect the AI model?

How could generative AI create new security challenges?

44. Suggested Classroom Activity

Activity: Design a Security Architecture

Ask groups to design security for:

A cloud-based university data analytics platform

They should identify:

Layer 1 — Network

Firewall

VPN

Secure communication

Layer 2 — Cloud

IAM

Shared responsibility



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EDUCATION TO INNOVATION



Encryption

Layer 3 — Big Data

Data access control

Distributed-system security

Audit logs

Layer 4 — AI

Training-data protection

Model security

Adversarial attack awareness

Layer 5 — Emerging Threats

Deepfakes

Generative AI risks

AI-enabled attacks